Staff Software Engineer (SWE) - Panel Interview Instructions

For the panel round of interviews, we would like you to:

1. **Read a bit more about Lore.** See, but do not share, "Resilience and Lore" at https://careers.lore.co/resilience-lore - password: lore-careers.
2. **Select one of the case studies to develop** that resonates with you and that you can use to show off your technical skillset. Let me know which one you select!
3. **Prepare a detailed brief of your solution with artifacts.** Demonstrate how you would approach solving the chosen case. This can be in the form of diagrams, flowcharts, code, tests, or any other visual or textual medium that conveys your approach to the problem and your suggested solution. Don't be afraid to go deep as you should expect to be challenged. Please be prepared to share your brief with each interviewer.
4. **Be prepared to discuss your approach in detail** during *each* interview, including the rationale behind your choices, any potential challenges you foresee, and how you would address them.

Case Studies

Select one of the case studies below to prepare.

We do not require a fully implemented solution, but rather a clear demonstration of your problem-solving skills and your ability to think critically about engineering challenges. You'll walk through your solution in three interviews. Since we work on cross-functional teams, each interviewer will bring a different lens--some more technical, some less. They'll be looking not just at your engineering skills, but also at how you operate in an environment that's asynchronous, collaborative, and interdisciplinary.

These scenarios are reflective of challenges you might encounter in the course of a role at Lore. We recognize the scenario cannot be sufficiently detailed in this brief, but we are unable to answer questions prior to the call. However, you are encouraged to make assumptions based on your experience to provide a solution strategy.

Case Study 1: Spark the Engagement

Scenario: Imagine you're working on an app designed to help people build meaningful micro-moments of reflection and emotional awareness throughout their day. The app already has a journaling feature and guided prompts to help users process what they wrote. However, engagement is inconsistent...some users log in daily, but others only show up sporadically.

Task: Design "Spark", a lightweight feature that provides a compelling reason for users to check in every day. The goal is to create something that integrates seamlessly with users' routines, sparks curiosity, and engenders emotional engagement without feeling like a chore or an intrusive notification.

As the developer behind this feature, you'll need to solve some key challenges:

- Users should be able to engage in under a minute.
- It shouldn't feel stale over time - you want meaningful engagement for your users.
- What would this look like? What would be most effective for the user experience you imagine?

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- What is the simplest demonstration of this you can build to show your idea?

Case Study 2: TrendForge's Community Pulse

Scenario: Imagine you're an engineer at TrendForge, a growing tech community platform where users discuss emerging technologies, AI, and programming languages. The platform has 10,000 active users and is steadily growing.

Task: Design Community Pulse, a feature that helps users stay informed about what's trending in a way that fosters engagement without overwhelming them. You need to decide what Community Pulse actually means...is that trending posts, most active users, hottest topics, sentiment analysis, or something else entirely?

As the developer behind this feature, you'll need to address a few key challenges:

- How do you surface insights without flooding users with too much data?
- While Trend's platform is small today, it's growing. How will your design hold up when the platform reaches 100,000+ users?
- What technical architecture would you recommend implementing this feature? What constraints would be most important?
- What is the simplest version of this you can show to your team about your hypothesis?

Case Study 3: Vulnerability Rules

Scenario: Imagine you're building a platform where people feel safe enough to be emotionally vulnerable on the internet. Vulnerability isn't something that scales easily. What feels intimate in a 1:1 conversation can feel performative or unsafe in a large group. However, we want to design a system where millions of people can engage in emotionally rich interactions at scale while maintaining a deep sense of psychological safety.

Task: Design a feature (or system) that enables Vulnerability at Scale. You'll need to solve problems around trust, moderation, and pacing.

As the owner of this feature, you'll need to address some key challenges:

- What fundamental principles would guide your design?
- How could you prevent emotional exhaustion, bad actors, and unintentional harm?
- What technical and UX guardrails could work to scale trust and psychological safety to millions of users?
- What is the simplest thing you can build to demonstrate how to figure out if your solution is trending in the right direction?

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